Useful spectral replacements and incidence stopping bounds for QC-GHP codes

24 October 2026 (Pacific) | Expanded stronger-bound follow-up to L-F16-stopping-bound

Abstract. Useful spectral replacements exist on explicit sparse QC-GHP families. Combining the classical fourth-moment cycle identity with the column-overlap bound yields stopping distance at least $\gamma+1$ whenever the degree-corrected fourth singular-value moment has zero excess. Explicit joint Sidon circulants satisfy the exact spectral equality, binary CSS commutation, fixed degrees and nontrivial logical content. The criterion needs neither the original second-eigenvalue gap nor a separate small-size exclusion. These ingredients are already available in classical literature and the motivating paper; we supply checked corollaries, explicit parameters and source-to-matrix bridges. Separately, a two-block incidence criterion gives $2(\gamma+\sigma)\leq(\sigma+\tau)|S|$, an explicit weight-four/eight family has exact stopping and logical distance six, and Gram-equal CSS twins prove an unbounded obstruction to uniformly informative Gram-only bounds. All new proofs are checked in Lean.

Question / source	Answer and scope
Abiad-Castriota,15 July 2026, v1,Prop. 5.1,Remark 5.2,§6: useful parameters for the stated bound	Answered negatively by the previous universal theorem; unchanged here.
Same source: tighter spectral stopping bounds, including multi-eigenvalue refinements	YES for useful-replacement existence: a fourth-moment criterion proves $\gamma+1$ on explicit sparse joint Sidon families. A separate Gram obstruction limits uniform improvement on other families.
This requested follow-up: incidence-conditioned replacement exceeding two on a sparse family beyond B1	YES. A reusable bound, explicit infinite families, nontrivial logical content, and a separate exact-six family.

The historical source question and this project-derived target are kept distinct. This paper adds no completed canonical author-question to the inventory. The proofs are published at an immutable source revision; all checked Lean files, including audit programs and dependencies, have full current and pinned URLs below.

1. A useful fourth-moment spectral replacement

Write $\eta_1,\dots,\eta_m$ for all ordered eigenvalues of the actual unnormalised real Gram matrix HH^T, counting multiplicity. If a binary incidence matrix has column degrees d_v and row degrees r_c, its fourth singular-value moment satisfies the zero-excess condition

$$\Sigma_i \eta_i^2 + \Sigma_v d_v = \Sigma_v d_v^2 + \Sigma_c r_c^2.$$

Theorem 1. Under this equality, if every column degree is at least γ, every nonempty stopping support has size at least $\gamma+1$. For a (γ,k)-regular matrix with N columns and m rows, the equality is $\Sigma_i \eta_i^2 + N\gamma = N\gamma^2 + mk^2$, equivalently $\Sigma_i \eta_i^2 = N\gamma(\gamma+k-1)$, using $mk=N\gamma$. No condition on η_2 and no presumed exclusion of small supports occurs.

Proof. The checked spectral theorem identifies $\Sigma \eta_i^2$ with $\mathrm{tr}((HH^T)^2)$, then with the sum of squared entries of H^TH. Let q_{uv} be the natural number of common checks of columns u,v. The diagonal is d_v, and $\Sigma_{uv} q_{uv} = \Sigma_c r_c^2$. The displayed condition is exactly $\Sigma_{u\neq v}(q_{uv}^2 - q_{uv}) = 0$. Each summand is nonnegative because q_{uv} is a natural number. Thus every distinct-column intersection is at most one. Covering the γ neighbors of a selected column by its intersections with the other selected columns gives $\gamma \le |S| - 1$. SpectralMoment.lean and MomentBound.lean check the ordered-eigenvalue identity, binary/cardinality bridges, finite sums, zero-excess implication and final stopping theorem.

This is a corollary of known ingredients: the motivating source [1], Lemma4.8, gives the cycle/moment identity, and Kashyap-Vardy [2], Theorem1, gives the overlap bound; Xia-Fu [8], Theorem3, also supplies the minimum-column-weight formulation. We do not claim a new classical counting principle. The precise existence interpretation of the §6 stronger-replacement direction is therefore positive; optimal universal refinements are a different target.

A globally valid replacement. For positive column weight γ, define L from $(m,N,\gamma,k,\Sigma\eta^2)$ to be $\gamma+1$ when the regular moment equality holds and2 otherwise. MomentReplacement.lean proves $L \ge 2$ and $L \le |S|$ for every nonempty stopping support of every positive-column-weight regular binary matrix. The fallback is proved from positive columns, rather than assuming away small supports. GlobalMomentFamily.lean proves $L=\gamma+1$, hence $L>2$ for $\gamma \ge 2$, on every constructed family member. Thus this is an actual bound function valid outside its informative regime too.

Quantitative excess extension. Let $\Delta = \Sigma\eta^2 + \Sigma d_v - \Sigma d_v^2 - \Sigma r_c^2$. For any natural μ with $\Delta < 2\mu(\mu+1)$, every stopping support satisfies $\gamma+\mu \le \mu|S|$ when the minimum column degree is γ. MomentExcess.lean proves this without regularity or an assumed intersection bound: an ordered distinct-column pair with overlap q contributes$2q(q-1)$ to the nonnegative total excess, forcing $q \le \mu$. The uniform covering theorem then applies. This supplies a checked criterion for small nonzero excess; it does not assert optimality or construct a new infinite family with prescribed nonzero excess.

1.1 Explicit QC-GHP parameters satisfying the replacement

Fix an even $\gamma \ge 2$ and any odd $n \ge 3^{2\gamma}$. Over Z/n take $S=\{3^i : 0 \le i < \gamma\}$ and $T=\{3^{\gamma+i} : 0 \le i < \gamma\}$. Let A and B be their binary circulants. Use the source scalar-factor convention

$$H_X = [A|B^T], \quad H_Z = [B|A^T].$$

The combined support $S \cup T$ is Sidon, including repeated summands, with no reduction of any relevant pair sum modulo n. The two supports are disjoint. Thus distinct columns meet in at

most one check. Circulants commute, giving $H_X H_Z^T = AB^T + B^T A = 0$ over F_2. There are n checks and 2n variables; each column has weight γ and each row weight2γ. The actual spectral equality is

$$\Sigma_i \eta_i^2 = 2n\gamma(3\gamma-1), \quad \text{hence } s(H_X) \geq \gamma+1.$$

The useful bound is proved again through Theorem1, with the constructed spectral identity as input. JointSidon.lean, JointMatrices.lean, JointPowers.lean and JointMomentFamily.lean check all hypotheses and the matrix/source bridge. Both binary kernel sectors also have nonzero weight at least γ+1 (JointKernel.lean). The all-ones-left word has zero X-syndrome and odd left parity, while every Z-stabilizer combination has even left parity; the code therefore has a nontrivial logical class. For fixed even γ, $n=3^{2\gamma}(2t+1)$ gives an infinite sparse family with fixed degrees. In particular γ=2 gives column/row weights2/4 and stopping guarantee3, beyond the earlier B1 family. No encoded-dimension formula or positive-rate claim is made.

1.2 Short support parameters and a54-variable certificate

Fixed degree4/8 family. Take S={0,1,3,11} and T={0,5,12,18} modulo any odd n≥37, and use $H_X=[A|B^T], H_Z=[B|A^T]$. Then N=2n, the column/row weights are4/8, $\Sigma\eta^2=88n$, and the globally valid spectral replacement equals5. Every nonempty X-stopping support and every nonzero kernel word in either CSS sector has weight at least5. The all-ones-left class is nontrivial. Thus this is a fixed-degree infinite sparse family starting at74 variables, with no exponential sufficient-length restriction. Exact stopping distance and an encoded-dimension formula are not claimed.

Proof and classical construction comparison. Each support separately is Sidon, and mixed sums s+t are unique with their ordered support roles. SeparateSidon.lean proves these weaker hypotheses suffice for column intersections≤1; disjoint supports and a Sidon union are unnecessary. ShortFamilySupports.lean kernel-checks the finite natural support identities and proves the no-wrap bridge for all n≥37, including cardinalities. Every relevant pair sum is at most36. ShortFamily.lean proves the actual binary CSS, exact moment, useful bound function, both-sector kernel bounds, nontrivial logical class and unbounded lengths. This is an explicit realization of the motivating source [1],Corollary4.11, which characterizes girth≥6 by combined nonzero difference multiplicities≤1. The source criterion and its classical design terminology receive credit.

Smaller finite instance. The same displayed supports also work modulo27, although the no-wrap sufficient condition does not apply. Practical27Core.lean checks every distinct-column overlap of the actual54-variable matrix in Lean's kernel. Practical27.lean gives $\Sigma\eta^2=2376$, stopping bound5 and a nontrivial logical class; Practical27Design.lean verifies the separate/mixed conditions and both-sector kernel bounds. This is an informative stopping-distance guarantee, not merely a spectral inequality. It is a lower bound, not an exact-five theorem. The finite search that suggested the supports is not a premise.

2. The two-block stopping bound

Let E(v,c) be a finite binary incidence relation. A stopping support S is nonempty and no check has exactly one selected neighbor. It need not be connected, elementary, or a stabilizer support. Partition the columns by a Boolean color. Suppose every selected column has at least γ neighbors, distinct selected columns of the same color have at most σ common neighbors, and columns of different colors have at most τ common neighbors, with σ≤τ. These hypotheses may be imposed globally to obtain a stopping-distance guarantee.

Theorem 2. Under these assumptions,

$$2(\gamma+\sigma) \le (\sigma+\tau)|S|.$$

If $\sigma+\tau>0$ this gives the integer bound $\operatorname{ceiling}(2(\gamma+\sigma)/(\sigma+\tau))$. The division-free statement also handles zero intersection limits. Both parts are allowed to be empty individually; nonemptiness is required only for S.

Proof. Fix a selected column v. Every check incident with v must meet another selected column. Covering its neighbors by the intersections with the other selected columns gives $\gamma \le \Sigma_{w \in S, w \neq v} |N(v) \cap N(w)|$. Let p and q count selected columns in the two parts. A selected left column therefore gives $\gamma+\sigma \le \sigma p+\tau q$; a selected right column gives $\gamma+\sigma \le \tau p+\sigma q$. If both parts occur, add these inequalities. If only one occurs, the corresponding inequality gives $\gamma+\sigma \le \sigma|S|$, which is at least as strong because $\tau \ge \sigma$. This proves the claim. All covering, finite-sum, partition and one-part cases are checked in PartitionBound.lean.

Classical comparison. Kashyap-Vardy [2], Theorem 1, assumes uniform column size w and maximum distinct-column intersection μ; its bound is $1+w/\mu$. The same count appears in the checked division-free theorem $\gamma+\mu \le \mu|S|$. Retaining the two parts refines this classical argument. At $\gamma=4, \sigma=1, \tau=2$, the uniform criterion gives three and Theorem 2 gives four. The exact-six result below uses additional finite incidence information.

3. Sidon supports give verified incidence hypotheses

For a finite abelian group G, call a support T Sidon when equality $a+b=c+d$ among its elements implies equality of the unordered pairs, including repeated elements. Let A be the binary circulant with $A_{c,j}=1$ exactly when $c-j \in T$, and set

$$H_X = H_Z = [A \mid A^T].$$

These are actual binary matrices with $|G|$ rows and $2|G|$ columns. Their two column neighborhoods are $j+T$ and $j-T$. A Sidon identity shows that distinct columns within one part share at most one check; across parts they share at most two. For the latter statement, after one common check c_0 is fixed, the only possible common checks are c_0 and $v+w-c_0$. The formal proof includes cardinality and empty-intersection cases. Every column has $|T|$ neighbors and every row has $2|T|$ neighbors.

Circulants commute with their transposes. Hence $H_X H_Z^T = AA^T + A^T A = 0$ over F_2. This is precisely the scalar QC-GHP/generalized-bicycle form of [1,4], with $r_1=n_1=1$ and $b=a$. Applying Theorem 2 gives $3|S| \ge 2(|T|+1)$, without any spectral or girth assumption.

4. An explicit sparse family for every fixed guarantee

Theorem 3. Fix an even integer $\gamma \ge 2$. For any odd $n \ge 3^\gamma$, use $G=Z/n$ and

$$T=\{1,3,3^2,\ldots,3^{\gamma-1}\}, \quad a(x)=\Sigma_{i=0}^{\gamma-1} x^{3^i},$$
$$H_X = H_Z = [A \mid A^T].$$

Then the matrices are binary CSS checks of column weight γ and row weight 2γ. Every nonempty stopping support has size at least $\operatorname{ceiling}(2(\gamma+1)/3)$. The quantum code has a nontrivial logical class. For fixed γ, $n=3^\gamma(2t+1), t \ge 0$, gives arbitrarily long matrices with the same degrees.

Proof of the support condition. Sort the exponents in $3^a+3^b=3^c+3^d$, with $a\le b$ and $c\le d$. If $b<d$, the left side is at most $2\cdot 3^b$, whereas the right side is at least $3^d\ge 3\cdot 3^b$, a contradiction. The opposite inequality is symmetric; cancel the equal largest terms to identify the smaller exponents. Every relevant pair sum is below n, since $2\cdot 3^{\gamma-1}<3^\gamma\le n$. Thus reduction modulo n preserves this Sidon property and all γ marks remain distinct. PowersThree.lean checks the inequalities, no-wrap cast bridge and cardinality; the support is not assumed Sidon as an unproved input.

Nontrivial quantum content. Take the binary word equal to one on every left variable and zero on the right. Its syndrome is zero because every row meets the left block in the even number γ. The sum of its left entries is one because n is odd. Every companion row has left sum zero, and so does every linear combination of companion rows. The word is therefore outside the stabilizer span. SidonMatrices.lean checks the finite-sum, binary-field and span-separation argument.

Fixed column / row weights	New guarantee / classical uniform guarantee
4 / 8	4 / 3
6 / 12	5 / 4
8 / 16	6 / 5
General even γ / 2γ	ceiling(2(γ+1)/3) / ceiling(γ/2+1)

Choosing γ first gives any desired fixed guarantee, and then length may increase with degrees fixed. This is not a claim of growing stopping distance at fixed degree, positive asymptotic rate, or practical parameter efficiency. The displayed sufficient length is exponential in γ. The known companion-row stopping mechanism gives a size 2γ upper witness; CSSStopping.lean checks that mechanism on the same binary matrices.

A second parameter choice makes the guarantee grow: $\gamma=2d$ and $n=3^{2d}$ give stopping and nonzero-kernel weight at least $d+1$. GrowingFamily.lean proves $(2d)3^d\le n$, so the incidence density is at most 3^{-d}. KernelFamily.lean supplies the binary/kernel bridge. This varies the degree with length; it does not assert growing stopping distance at fixed degree.

5. A separate family with exact stopping and logical distance six

Theorem 4. For every positive integer t, let $n=39t$ and use $a(x)=1+x+x^4+x^6$ with $H_X=H_Z=[A|A^T]$. There are $78t$ variables and $39t$ checks, column weight 4 and row weight 8. The stopping distance and the quantum logical distance are both exactly six.

The reusable intersection criterion proves four for this entire family. The stronger exact result is certified independently: simultaneous coordinate translation reduces the base $n=39$ search to the two side representatives. A sound degree-one extension procedure rejects both roots with four further additions, excluding all stopping supports of size at most five. Its exact Boolean tests are reduced by Lean’s kernel. No external search result is a premise.

Reduction modulo 39 maps lifted column lists onto their base lists, lifts every incident base check, and is injective on each check’s neighbor set. The checked local-cover theorem therefore projects a stopping support to a base stopping support of no greater size. This transfers the lower bound six to every positive 39-multiple lift; it is not extrapolation from sampled lengths.

A six-variable upper witness is

$$\text{left}(0),\ \text{left}(-5),\ \text{left}(-4),$$
$$\text{right}(1),\ \text{right}(5),\ \text{right}(6).$$

The six variables are distinct at every stated lift. Every incident check has a selected partner. A separate exact parity calculation shows that every selected check degree is even; the binary word is in the kernel. It has left weight three, whereas each companion row has left weight four. The same parity functional excludes every stabilizer combination. Every nonzero binary kernel word has stopping support, so the lower stopping bound gives logical weight at least six; the nontrivial six-word gives equality in both CSS sectors, which coincide here. Unlike the completed B1 family, this upper witness is a logical word rather than a companion row. It is also not assumed elementary: the independent base reconstruction has selected degrees zero, two and four.

BicycleMatrices.lean proves the actual circulant/transpose binary matrix entries equivalent to the incidence relation and checks CSS commutation. BicycleDegrees.lean proves dimensions and degrees. The translation, kernel certificate, cover, cardinality, syndrome and stabilizer-span bridges are all included in the proof chain. The statement does not assert a formula for the logical dimension. Independent binary elimination gives $k=6$ for the first three sampled lifts; that is a computation recorded separately, not a new general rank theorem.

6. What the full real Gram matrix cannot certify

Theorem 5. For the same constituent circulant A, define

$$\text{Good:}\quad H_X=H_Z=[A|A^T];$$
$$\text{Bad:}\quad H'_X=[A|A],\quad H'_Z=[A^T|A^T].$$

Both pairs are binary CSS checks whose X-matrices have the same numbers of rows and columns and the same row/column degrees. No equality of encoded quantum dimension is assumed or claimed. Over the reals,

$$H_X H_X{}^T=AA^T+A^TA=2AA^T=H'_X H'_X{}^T.$$

The bad X-matrix has two identical positive columns at each coordinate, yielding a stopping pair. Positive columns exclude singletons, so its stopping distance is exactly two. The good family of Theorem 4 has distance six. More generally the same identity holds for every support in Theorem 3, together with a bad stopping pair.

Let L be any proposed universally valid real-valued lower-bound function whose inputs consist only of this real Gram matrix and the shared check-matrix dimensions/degrees. Validity on the bad counterpart forces $L\le 2$; identical input data force the same output on the good code. GramObstruction.lean, BadStopping.lean, SidonBad.lean and RealGramBound.lean prove this statement against the actual matrices and stopping supports. Any function of the complete Gram spectrum, or of eigenvectors determined solely by that Gram matrix, is included. An additional incidence hypothesis may distinguish the codes and is precisely what the replacement uses.

Unbounded factor loss. Fix $M\ge 1$ and use the paired powers family with $\gamma=4M$ and $n=3^{4M}(2t+1)$. Every good stopping support has size at least$2M+1$; the bad twin has stopping distance2. Therefore any real-valued Gram-only lower bound valid on the bad member satisfies $M\cdot L<|S|$ for every good stopping support. GramGap.lean checks this statement with the exact matrices and all quantifiers. The bad weight-two word is outside the companion

stabilizer span; BadLogical.lean checks both sector lower bounds and the nontrivial logical witness. These are not zero-information quantum examples. Binary rank and encoded dimension remain additional data outside the obstruction.

BothGram.lean further proves equality of the two separate real check Gram matrices and the corresponding bound obstruction. This does not include cross-sector real incidence data, the variable-side eigenvectors of the original matrices, or binary ranks.

7. Exact comparison with spectral and known stopping bounds

The motivating source [1], Proposition 5.1, requires $\sigma_2(H_\chi)^2<\gamma/2$ and separately excludes the small root branch before concluding a stopping-distance lower bound. The preceding universal parameter theorem answers that proposition's useful-parameter question negatively in its stated scalar-factor model. That answer is preserved. Theorem 2 replaces its spectral step with incidence information; it does not claim parameters satisfying the source inequality.

Zhu-Chugg [3], Corollary 6, equations 41-42, give the two regular Tanner-graph expressions below, with N variables, column degree γ, row degree k and second Gram eigenvalue η:

$$N(2\gamma-\eta)/(\gamma k-\eta),$$
$$2N(2\gamma+k-2-\eta)/(k(\gamma k-\eta)).$$

Their regular connected setting and strict gap $\eta<\gamma k$ must be respected. On every39-cover of the exact-six family, an actual two-dimensional rational Rayleigh certificate proves $\eta\geq290/13$. Its base vectors have disjoint supports, squared norms13, energies332 and cross energy42. The minimum plane quotient is $(332-42)/13$. SpectralPlane.lean proves the ordered-second-eigenvalue conclusion; FiberSums.lean and BicycleCoverEnergy.lean prove exact norm/energy scaling through the cyclic cover; BicycleSpectralFamily.lean proves the family result. Consequently both Zhu-Chugg expressions are strictly negative whenever $\eta<32$. If the gap fails, those formulas are outside their stipulated regime. No floating-point estimate is a proof premise.

On the new joint-support instance $\gamma=2$,$n=81$, the fourth-moment certificate is1620 and Theorem1 gives3. JointSpectralBase.lean checks another rational plane certificate: squared norms40, energies280 and cross energy38 give $\eta\geq121/20>6$. Under $\eta<8$, both Zhu-Chugg expressions are strictly negative. Thus the displayed spectral replacement is informative on a concrete instance where those two formulas are not. Other applicability hypotheses of the cited theorem are not discarded by this algebraic comparison.

The54-variable instance has another actual ordered-second-eigenvalue certificate: $u=\delta_0+\delta_8-\delta_4-\delta_{12}$ and $v=\delta_{20}+\delta_1-\delta_{24}-\delta_5$ have disjoint supports, squared norms4, energies34 and cross energy0, proving $\eta\geq17/2$. Under $\eta<32$, the first Zhu-Chugg expression is negative, and the second is at most$297/94<4$, whereas the moment replacement gives5. Practical27Spectral.lean checks all integer evaluations, real casts, the eigenvalue conclusion and both formula comparisons. This preserves the cited theorem's other applicability hypotheses.

For illustration only, independent reconstruction gives $\eta\approx27.52509838$ for $n=39$, and $\eta\approx7.53208889$ for the joint $n=81$ code. Exact integer checks cover eight39-lifts, twelve joint-family parameter choices and all certificate numerators. Sampled ranks and floating-point eigenvalues remain diagnostics, separate from every theorem.

Kashyap-Vardy [2] supplies the classical uniform-overlap criterion already sufficient for a positive answer exceeding two in the weight-four case. Theorem 2 improves its value by retaining the block partition. Schwartz-Vardy [5] emphasizes that stopping distance belongs to a specified parity-check matrix and that codeword supports are stopping sets. Their stopping redundancy instead optimizes over added checks; that problem is different from our fixed-H_X guarantee. The generic binary kernel/stopping bridge is checked here.

Wang-Pryadko [4], Eq.8, uses $H_X=[A|B]$,$H_Z^{\top}=[B;-A]$; our paired construction takes $B=A^{\top}$. Their distance bounds and gcd dimension formula concern logical code parameters. A logical-distance lower bound alone does not exclude non-codeword stopping supports. Kuo-Ouyang [6] studies decoders exploiting degeneracy; success of such a decoder does not establish a lower bound for ordinary peeling stopping sets. The known stabilizer upper mechanism and the new nontrivial logical upper witness are distinguished above.

Dehghan-Banihashemi [9] gives earlier cospectral bipartite examples with matched degree sequences and different cycle counts. Their inspected cycle statements explain why spectral information can omit incidence data; they are not the actual full-Gram-equal binary CSS family and stopping/logical certificates proved here. The existing F38 full-text screen and its two recorded directions are reused, without inventing a new inventory cohort.

Xia-Fu [8], Theorem3, assumes minimum column weight γ, maximum distinct-column intersection λ and integral γ/λ. It gives the pseudo-weight bound $\gamma/\lambda+1$ and an equality characterization. At $\gamma=4,\lambda=2$ it gives3, whereas the two-block refinement gives4. Its girth$\geq$6 Theorem1 is inapplicable to the paired family’s cross-block4-cycles. These hypotheses, pseudo-weight versus stopping distance, and the two matrix families are kept distinct.

Public GitHub comparisons include the pinned bicycle-race README, which describes logical-distance records, and the Kuo decoder repository metadata. A final CLI support-fingerprint query and a known-positive mathlib control both returned no items; those negative code-search outputs are discarded. Authenticated Edge access later succeeded: the arXiv-ID query visibly returned the source announcement and an earlier project lift-label post. The source post detail displayed10:23PM,15July2026; the prior UTC16July date is distinguished in the chronology. A focused QC-GHP stopping query displayed no results in that bounded view. None is a proof of global absence or priority. The earlier Mac-locked failure is superseded by this fresh read-only check. No post was sent.

The post-lemma audit also inspected Donovan-Rao-Üsküplü-Yazıcı [10], whose Lemma9 gives stopping distance$\geq$8 when3 does not divide a for their specified DM/DCA parity-check matrices, with four disjoint row groups and the source construction hypotheses. Those classical matrices and their stopping patterns are compared as stated; no QC-GHP companion/CSS bridge is supplied by that result for our displayed matrices. Chee-Ling-Yin [11] distinguishes cyclic difference packing from the additional partition requirement of PCDP; our overlapping pair of supports is not a claimed PCDP. Gholami-Samadieh [12] studies classical lifted finite set systems and inevitable walks. Amirzade-Sadeghi [13] gives classical single/multiple-edge difference criteria and lifting-degree bounds; it supplies no stated QC-GHP stopping or CSS-companion theorem for these matrices. These are concrete construction comparisons, not evidence that classical stopping bounds above two are new.

8. Evidence, inventory and remaining work

The aggregate build and all 485 named-declaration audits pass in Lean4.36.0-rc1 with mathlib5f7b4198d22a9c82e1c698a379334fe128095c82. Only propext,Classical.choice and Quot.sound occur. The project contains 50 new modules and four frozen imported research modules. Reproduce with lake build StrongerAnswer, followed by lake env lean AuditStronger.lean. The logs, signatures, complete source links, hashes and independent reconstructions accompany the archive.

Proof component	Checked source
Ordered eigenvalue fourth moment, zero excess and useful spectral bound	SpectralMoment; MomentBound
Globally valid bound function and quantitative nonzero excess	MomentReplacement; GlobalMomentFamily; MomentExcess
Shorter fixed-degree family and54-variable matrix, both sectors	SeparateSidon; ShortFamilySupports; ShortFamily; Practical27Core; Practical27; Practical27Design; Practical27Spectral
Actual source matrices, length/density consumers and bad companion degrees	SourceConsumers; BadDegrees
Constructed joint Sidon matrices, spectral family and both binary sectors	JointSidon; JointMatrices; JointPowers; JointMomentFamily; JointKernel
Exact plane certificates and all39-cover energy scaling	SpectralPlane; FiberSums; BicycleSpectralBase; BicycleCoverEnergy; BicycleSpectralFamily; JointSpectralBase
Real-valued/unbounded Gram gap and nontrivial bad logical class	RealGramBound; GramGap; BadLogical; BothGram
Growing-degree density and kernel guarantees	GrowingFamily; KernelFamily
Generic neighbor cover and two-block/uniform bounds	PartitionBound.lean
Sidon intersection limits and no-wrap powers-of-three support	SidonStopping.lean; PowersThree.lean
Binary matrices, CSS, degree counts and logical separation	SidonMatrices.lean; PowersFamily.lean
Rooted search soundness, two roots and cyclic covers	dependencies/IncidenceCore,List,Cover; BicycleBaseCore,Base,FamilyCore,Family
Concrete actual matrices, degrees, overlap and exact logical witness	BicycleMatrices,Degrees,Overlap,Logical; KernelStopping
Same-Gram identity, bad pair and universal-bound obstruction	GramObstruction; BadStopping; SidonBad
Known nonzero companion-row stopping mechanism	CSSStopping.lean
Consumers and exhaustive named-declaration audit	StrongerAnswer.lean; AuditStronger.lean

Nine additional accessible main texts were screened, adding28 historical questions/directions, nine precise enough for the denominator:267→295 entries,135→144 eligible targets,110→119 papers and97→106 full-text main-paper screens. The separate indexing deltas are20/8/4,3/1/1

and5/0/4. Completed canonical Lean targets remain five. New harvested questions have unassessed current resolution, Lean discovery and effort. The latest four-paper target screen records five broad directions; FSS and the lifting-degree paper have no additional explicit unresolved target. The PCDP theorem’s excluded residue classes are not invented as an open conjecture: explicit8/16/32seeds exist in the same paper. Existing F38 screening is reused without duplication. Publisher429/403, supplementary/version, retrieval and X access gaps remain explicit.

Question answered. Yes: useful spectral-replacement existence is proved by the degree-corrected fourth moment on an explicit sparse family; useful incidence-conditioned replacement existence is also proved beyond B1. The original Proposition5.1 useful-parameter question remains answered negatively. **Limit.** The spectral obstruction is an unbounded family result for the specified shared data, not universal impossibility on all other classes. **Next bounded follow-up.** Construct or obstruct sparse scalar QC families in the useful nonzero-excess regime of the checked quantitative criterion, or determine sharp stopping distances for the general powers families. Neither is required to finish the positive existence answers.

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[13] Farzane Amirzade,Mohammad-Reza Sadeghi. Lower bounds on the lifting degree of single-edge and multiple-edge QC-LDPC codes by difference matrices. arXiv:1709.00825v1,4September2017; internal running draft date18November2018. Full-document target screen, no additional explicit unresolved target. https://arxiv.org/abs/1709.00825v1

Appendix A. Actual checked Lean statements and proofs

The following excerpts are verbatim from the immutable checked sources. They expose the quantifiers and decisive proof steps. The complete dependency guide follows, and every file is linked individually in AppendixB. No numerical spectral premise is substituted for an actual matrix statement.

PartitionBound.lean

```
theorem partition_bound (color : V → Bool) (S : Finset V)
    (hS : Incidence.Stopping E S) (γ σ τ : ℕ) (hst : σ ≤ τ)
    (hdeg : ∀ v ∈ S, γ ≤ (neighbors E v).card)
    (hov : ∀ v ∈ S, ∀ w ∈ S, v ≠ w →
      overlap E v w ≤ if color v = color w then σ else τ) :
    2 * (γ + σ) ≤ (σ + τ) * S.card := by
  classical
  let P := S.filter (fun w => color w = false)
  let Q := S.filter (fun w => color w = true)
  have hcard : P.card + Q.card = S.card := by
    have h := card_filter_add_card_filter_not (s := S) (fun w => color w = false)
    simpa [P,Q,Bool.not_eq_false] using h
  have hfalse (v : V) (hv : v ∈ S) (hc : color v = false) :
      γ + σ ≤ σ * P.card + τ * Q.card := by
    have h := column_budget E color S hS γ σ τ hdeg hov v hv
    simpa [hc,P,Q,Bool.not_eq_false] using h
  have htrue (v : V) (hv : v ∈ S) (hc : color v = true) :
      γ + σ ≤ τ * P.card + σ * Q.card := by
    have h := column_budget E color S hS γ σ τ hdeg hov v hv
    simpa [hc,P,Q,Bool.not_eq_true,add_comm] using h
  by_cases hp : P.Nonempty
  · obtain ⟨v,hv⟩ := hp
    have hv' := mem_filter.mp hv
    have hb := hfalse v hv'.1 hv'.2
    by_cases hq : Q.Nonempty
    · obtain ⟨w,hw⟩ := hq
      have hw' := mem_filter.mp hw
      have hb' := htrue w hw'.1 hw'.2
      nlinarith
    · have hq0 : Q.card = 0 := card_eq_zero.mpr (not_nonempty_iff_eq_empty.mp hq)
      nlinarith [Nat.mul_le_mul_right P.card hst]
  · have hp0 : P.card = 0 := card_eq_zero.mpr (not_nonempty_iff_eq_empty.mp hp)
    obtain ⟨v,hv⟩ := hS.1
    have hc : color v = true := by
      cases hc : color v
      · have : v ∈ P := mem_filter.mpr ⟨hv,hc⟩
        exact False.elim (hp ⟨v,this⟩)
      · rfl
    have hb := htrue v hv hc
    nlinarith [Nat.mul_le_mul_right Q.card hst]
```

PowersFamily.lean

```
theorem positive_family (γ n : ℕ) [NeZero n]
    (hn : 3^γ ≤ n) (heven : Even γ) (hodd : Odd n) :
    let S := support γ n
    (SidonStopping.X (ZMod 2) S * (SidonStopping.X (ZMod 2) S)ᵀ = 0) ∧
    (∀ v, (Finset.univ.filter (fun c => SidonStopping.X (ZMod 2) S c v = 1)).card = γ) ∧
    (∀ c, (Finset.univ.filter (fun v => SidonStopping.X (ZMod 2) S c v = 1)).card = 2*γ) ∧
    (∀ T, Incidence.Stopping (SidonStopping.E S) T → 2*(γ+1) ≤ 3*T.card) ∧
    (SidonStopping.X (ZMod 2) S).mulVec (SidonStopping.leftOnes (G := ZMod n)) = 0 ∧
    (¬ ∃ a, (SidonStopping.X (ZMod 2) S)ᵀ.mulVec a =
      SidonStopping.leftOnes (G := ZMod n)) ∧
    (SidonStopping.X ℝ S * (SidonStopping.X ℝ S)ᵀ =
      SidonStopping.BadX ℝ S * (SidonStopping.BadX ℝ S)ᵀ) := by
  dsimp only
  have hcard := support_card γ n hn
  have he : Even (support γ n).card := by simpa [hcard] using heven
  refine ⟨SidonStopping.css _,?_,?_,?_,SidonStopping.leftOnes_kernel _ he,?_,SidonStopping.gram_equal _⟩
  · intro v; rw [SidonStopping.column_weight,hcard]
  · intro c; rw [SidonStopping.row_weight,hcard]
  · intro T hT
    simpa only [hcard] using SidonStopping.stopping_bound _ (sidon γ n hn) T hT
  · apply SidonStopping.leftOnes_not_stabilizer _ he
    simpa using hodd
```

KernelStopping.lean

```
theorem quantum_distance_six (h : 39 | n) :
    (HX (n := n) (ZMod 2) * (HZ (n := n) (ZMod 2))ᵀ = 0) ∧
    (∀ x : Var n → ZMod 2, x ≠ 0 → (HX (ZMod 2)).mulVec x = 0 →
      6 ≤ (BinaryStopping.support x).card) ∧
    (HX (ZMod 2)).mulVec (@word n) = 0 ∧
    (¬ ∃ a : Check n → ZMod 2, ((HZ (ZMod 2))ᵀ.mulVec a) = (@word n)) ∧
    (BinaryStopping.support (@word n)).card = 6 :=
  ⟨css,all_nonzero_kernel_weight_ge_six h,word_kernel h,word_not_stabilizer h,word_weight h⟩
```

SidonBad.lean

```
theorem gram_only_bound (S : Finset G)
    (lower : Matrix G G ℝ → ℕ)
    (valid_bad : ∀ T : Finset (Bool × G), Incidence.Stopping (badE S) T →
      lower (BadX ℝ S * (BadX ℝ S)ᵀ) ≤ T.card) :
    lower (X ℝ S * (X ℝ S)ᵀ) ≤ 2 := by
  rw [gram_equal]
  have h := valid_bad twin (twin_stopping S)
  simpa [twin_card] using h
```

GramGap.lean

```
theorem unbounded_gram_gap (M t : ℕ) [NeZero (liftLength (2*M) t)]
    (lower : Matrix (ZMod (liftLength (2*M) t)) (ZMod (liftLength (2*M) t)) ℝ → ℝ)
    (valid_bad : ∀ U : Finset (Bool × ZMod (liftLength (2*M) t)),
      Incidence.Stopping (SidonStopping.badE (support (4*M) (liftLength (2*M) t))) U →
      lower (SidonStopping.BadX ℝ (support (4*M) (liftLength (2*M) t)) *
        (SidonStopping.BadX ℝ (support (4*M) (liftLength (2*M) t)))ᵀ) ≤ (U.card : ℝ))
    (T : Finset (Bool × ZMod (liftLength (2*M) t)))
    (hT : Incidence.Stopping (SidonStopping.E (support (4*M) (liftLength (2*M) t))) T) :
    (M:ℝ)*lower (SidonStopping.X ℝ (support (4*M) (liftLength (2*M) t)) *
      (SidonStopping.X ℝ (support (4*M) (liftLength (2*M) t)))ᵀ) < (T.card : ℝ) := by
  have hbound := SidonStopping.real_gram_only_bound _ lower valid_bad
  have hrewrite : 4*M = 2*(2*M) := by omega
  rw [hrewrite] at hT
  have hg := arbitrary_fixed_guarantee (2*M) t T hT
  have hgr : (2*M+1:ℕ) ≤ T.card := hg
  have hrr : (2*(M:ℝ)+1) ≤ (T.card:ℝ) := by exact_mod_cast hgr
  have hmul := mul_le_mul_of_nonneg_left hbound (Nat.cast_nonneg M : (0:ℝ) ≤ M)
  linarith
```

BicycleSpectralFamily.lean

```
theorem zhu_chugg_negative (h : 39 | n)
    (hgap : eigs (Family.HX (n := n) ℝ) ⟨1,by exact check_count h⟩ < 32) :
    let s := eigs (Family.HX (n := n) ℝ) ⟨1,by exact check_count h⟩
    (2*(n:ℝ))*(8-s)/(32-s) < 0 ∧
    (4*(n:ℝ))*(14-s)/(8*(32-s)) < 0 := by
  dsimp
  have hs := second_lower h
  have hn : (0:ℝ) < n := by exact_mod_cast NeZero.pos n
  constructor
  · apply div_neg_of_neg_of_pos
    · exact mul_neg_of_pos_of_neg (by positivity) (by linarith)
    · linarith
  · apply div_neg_of_neg_of_pos
    · exact mul_neg_of_pos_of_neg (by positivity) (by linarith)
    · positivity
```

MomentBound.lean

```
theorem regular_spectral_bound (E : V → C → Prop) [DecidableRel E] (γ k : ℕ)
    (hcol : ∀ v, (PartitionStopping.neighbors E v).card = γ)
    (hrow : ∀ c, rowWeight E c = k)
    (hmoment : (∑ i,(StoppingSpectral.eigs (H E) i)^2) + (Fintype.card V : ℝ)*γ =
      (Fintype.card V : ℝ)*(γ:ℝ)^2 + (Fintype.card C : ℝ)*(k:ℝ)^2)
    (S : Finset V) (hS : Incidence.Stopping E S) : γ+1 ≤ S.card := by
  apply spectral_stopping_bound E γ (fun v => by rw [hcol]) _ S hS
  simpa only [hcol,hrow,sum_const,card_univ,nsmul_eq_mul] using hmoment
```

JointMomentFamily.lean

```
theorem fourth_moment (hn : 3^(2*γ) ≤ n) :
    (∑ i,(eigs (JointSidon.X ℝ (leftMarks γ n) (rightMarks γ n)) i)^2) =
      2*(n:ℝ)*(γ:ℝ)*(3*(γ:ℝ)-1) := by
  have hm := StoppingMoment.moment_of_overlap_one
    (JointSidon.E (leftMarks γ n) (rightMarks γ n))
    (JointSidon.overlap_one _ _ (joint_sidon γ n hn) (disjoint_marks γ n hn))
  rw [real_matrix] at hm
  simp_rw [column_degree hn,row_degree hn] at hm
  simp only [sum_const,card_univ,Fintype.card_prod,Fintype.card_bool,ZMod.card,
    Nat.cast_mul,Nat.cast_ofNat,nsmul_eq_mul] at hm
  nlinarith
```

ShortFamily.lean

```
theorem source_stopping_five (hn : 37≤n) (U : Finset (Bool × ZMod n))
    (hU : Incidence.Stopping (fun v c => HX (n := n) (ZMod 2) c v=1) U) : 5≤U.card := by
  refine regular_spectral_bound (E (n := n)) 4 8 (degree hn) (row_degree hn) ?_ U ?_
  · rw [real_matrix,fourth_moment hn]
    simp only [Fintype.card_prod,Fintype.card_bool,ZMod.card,Nat.cast_mul,Nat.cast_ofNat]
    ring
  · simpa only [HX,Incidence.Stopping,JointSidon.binary_incidence] using hU

theorem replacement_value (hn : 37≤n) :
    regularReplacement n (2*n) 4 8 (∑ i,(eigs (HX (n := n) ℝ) i)^2)=5 := by
  rw [fourth_moment hn]
  unfold regularReplacement
  split_ifs with he
  · norm_num
  · exfalso
    apply he
    push_cast
    ring
```

Practical27Spectral.lean

```
theorem zhu_chugg_comparison (hgap : eigs HR ⟨1,by decide⟩ < 32) :
    let s := eigs HR ⟨1,by decide⟩
    54*(8-s)/(32-s) < 0 ∧
      108*(14-s)/(8*(32-s)) ≤ (297:ℝ)/94 ∧ (297:ℝ)/94 < 4 := by
  dsimp
  have hs := second_lower
  refine ⟨?_,?_,by norm_num⟩
  · apply div_neg_of_neg_of_pos
    · nlinarith
    · linarith
  · apply (div_le_iff₀ (by positivity : 0 < 8*(32-eigs HR ⟨1,by decide⟩))).mpr
    nlinarith
```

Dependency guide

PartitionBound provides generic counting. SidonStopping and PowersThree prove incidence hypotheses constructively. SidonMatrices and PowersFamily supply actual binary CSS and nontrivial logical content. BicycleBaseCore/List dependencies give search soundness; BicycleBase rejects small roots; FamilyCore/Cover transfer exact6. BicycleMatrices/Degrees/Logical/KernelStopping check the actual code and logical witness. GramObstruction, BadStopping and SidonBad prove indistinguishability and exact bad stopping2. CSSStopping checks the known stabilizer upper mechanism. AuditStronger checks every named declaration, including every frozen dependency. The second checkpoint adds SpectralMoment/MomentBound for an actual ordered-eigenvalue replacement; JointSidon/Matrices/Powers/MomentFamily/Kernel for constructed useful spectral families; JointSpectralBase for an exact cited-formula comparison; real-valued/unbounded Gram-gap and nontrivial bad logical proofs; exact two-vector certificates with cyclic-cover energy transfer; and growing stopping/kernel guarantees. SeparateSidon/ShortFamilySupports/ShortFamily provide shorter fixed-degree parameters; Practical27Core/Practical27/Practical27Design/Practical27Spectral check the54-variable example and exact cited-formula comparison. MomentReplacement/GlobalMomentFamily and MomentExcess provide the global bound function and quantitative extension. BothGram covers both separate check Gram matrices, excluding cross-sector/binary-rank information.

Appendix B. Full Lean URLs

All checked Lean files are included, with current and immutable clickable URLs. The immutable revision is bed7719cb3cac4118e4f333690db364566ede03e. The audit itself and all inherited research dependencies are listed.

AuditStronger.lean - axiom/signature audit

Current: https://github.com/alectionary/vibe-math/blob/master/formalization/stopping-bound/stronger/AuditStronger.lean

Immutable: https://github.com/alectionary/vibe-math/blob/bed7719cb3cac4118e4f333690db364566ede03e/formalization/stopping-bound/stronger/AuditStronger.lean

BadDegrees.lean - new checked proof module

Current: https://github.com/alectionary/vibe-math/blob/master/formalization/stopping-bound/stronger/BadDegrees.lean

Immutable: https://github.com/alectionary/vibe-math/blob/bed7719cb3cac4118e4f333690db364566ede03e/formalization/stopping-bound/stronger/BadDegrees.lean

BadLogical.lean - new checked proof module

Current: https://github.com/alectionary/vibe-math/blob/master/formalization/stopping-bound/stronger/BadLogical.lean

Immutable: https://github.com/alectionary/vibe-math/blob/bed7719cb3cac4118e4f333690db364566ede03e/formalization/stopping-bound/stronger/BadLogical.lean

BadStopping.lean - new checked proof module

Current: https://github.com/alectionary/vibe-math/blob/master/formalization/stopping-bound/stronger/BadStopping.lean

Immutable: https://github.com/alectionary/vibe-math/blob/bed7719cb3cac4118e4f333690db364566ede03e/formalization/stopping-bound/stronger/BadStopping.lean

BicycleBase.lean - new checked proof module

Current: https://github.com/alectionary/vibe-math/blob/master/formalization/stopping-bound/stronger/BicycleBase.lean

Immutable: https://github.com/alectionary/vibe-math/blob/bed7719cb3cac4118e4f333690db364566ede03e/formalization/stopping-bound/stronger/BicycleBase.lean

BicycleBaseCore.lean - new checked proof module

Current: https://github.com/alectionary/vibe-math/blob/master/formalization/stopping-bound/stronger/BicycleBaseCore.lean

Immutable: https://github.com/alectionary/vibe-math/blob/bed7719cb3cac4118e4f333690db364566ede03e/formalization/stopping-bound/stronger/BicycleBaseCore.lean

BicycleCoverEnergy.lean - new checked proof module

Current: https://github.com/alectionary/vibe-math/blob/master/formalization/stopping-bound/stronger/BicycleCoverEnergy.lean

Immutable: https://github.com/alectionary/vibe-math/blob/bed7719cb3cac4118e4f333690db364566ede03e/formalization/stopping-bound/stronger/BicycleCoverEnergy.lean

BicycleDegrees.lean - new checked proof module

Current: https://github.com/alectionary/vibe-math/blob/master/formalization/stopping-bound/stronger/BicycleDegrees.lean

Immutable: https://github.com/alectionary/vibe-math/blob/bed7719cb3cac4118e4f333690db364566ede03e/formalization/stopping-bound/stronger/BicycleDegrees.lean

Appendix B. Full Lean URLs (continued)

BicycleFamily.lean - new checked proof module

Current: https://github.com/alectionary/vibe-math/blob/master/formalization/stopping-bound/stronger/BicycleFamily.lean

Immutable: https://github.com/alectionary/vibe-math/blob/bed7719cb3cac4118e4f333690db364566ede03e/formalization/stopping-bound/stronger/BicycleFamily.lean

BicycleFamilyCore.lean - new checked proof module

Current: https://github.com/alectionary/vibe-math/blob/master/formalization/stopping-bound/stronger/BicycleFamilyCore.lean

Immutable: https://github.com/alectionary/vibe-math/blob/bed7719cb3cac4118e4f333690db364566ede03e/formalization/stopping-bound/stronger/BicycleFamilyCore.lean

BicycleLogical.lean - new checked proof module

Current: https://github.com/alectionary/vibe-math/blob/master/formalization/stopping-bound/stronger/BicycleLogical.lean

Immutable: https://github.com/alectionary/vibe-math/blob/bed7719cb3cac4118e4f333690db364566ede03e/formalization/stopping-bound/stronger/BicycleLogical.lean

BicycleMatrices.lean - new checked proof module

Current: https://github.com/alectionary/vibe-math/blob/master/formalization/stopping-bound/stronger/BicycleMatrices.lean

Immutable: https://github.com/alectionary/vibe-math/blob/bed7719cb3cac4118e4f333690db364566ede03e/formalization/stopping-bound/stronger/BicycleMatrices.lean

BicycleOverlap.lean - new checked proof module

Current: https://github.com/alectionary/vibe-math/blob/master/formalization/stopping-bound/stronger/BicycleOverlap.lean

Immutable: https://github.com/alectionary/vibe-math/blob/bed7719cb3cac4118e4f333690db364566ede03e/formalization/stopping-bound/stronger/BicycleOverlap.lean

BicycleSpectralBase.lean - new checked proof module

Current: https://github.com/alectionary/vibe-math/blob/master/formalization/stopping-bound/stronger/BicycleSpectralBase.lean

Immutable: https://github.com/alectionary/vibe-math/blob/bed7719cb3cac4118e4f333690db364566ede03e/formalization/stopping-bound/stronger/BicycleSpectralBase.lean

BicycleSpectralFamily.lean - new checked proof module

Current: https://github.com/alectionary/vibe-math/blob/master/formalization/stopping-bound/stronger/BicycleSpectralFamily.lean

Immutable: https://github.com/alectionary/vibe-math/blob/bed7719cb3cac4118e4f333690db364566ede03e/formalization/stopping-bound/stronger/BicycleSpectralFamily.lean

BothGram.lean - new checked proof module

Current: https://github.com/alectionary/vibe-math/blob/master/formalization/stopping-bound/stronger/BothGram.lean

Immutable: https://github.com/alectionary/vibe-math/blob/bed7719cb3cac4118e4f333690db364566ede03e/formalization/stopping-bound/stronger/BothGram.lean

Appendix B. Full Lean URLs (continued)

CSSStopping.lean - new checked proof module

Current: https://github.com/alectionary/vibe-math/blob/master/formalization/stopping-bound/stronger/CSSStopping.lean

Immutable: https://github.com/alectionary/vibe-math/blob/bed7719cb3cac4118e4f333690db364566ede03e/formalization/stopping-bound/stronger/CSSStopping.lean

FiberSums.lean - new checked proof module

Current: https://github.com/alectionary/vibe-math/blob/master/formalization/stopping-bound/stronger/FiberSums.lean

Immutable: https://github.com/alectionary/vibe-math/blob/bed7719cb3cac4118e4f333690db364566ede03e/formalization/stopping-bound/stronger/FiberSums.lean

GlobalMomentFamily.lean - new checked proof module

Current: https://github.com/alectionary/vibe-math/blob/master/formalization/stopping-bound/stronger/GlobalMomentFamily.lean

Immutable: https://github.com/alectionary/vibe-math/blob/bed7719cb3cac4118e4f333690db364566ede03e/formalization/stopping-bound/stronger/GlobalMomentFamily.lean

GramGap.lean - new checked proof module

Current: https://github.com/alectionary/vibe-math/blob/master/formalization/stopping-bound/stronger/GramGap.lean

Immutable: https://github.com/alectionary/vibe-math/blob/bed7719cb3cac4118e4f333690db364566ede03e/formalization/stopping-bound/stronger/GramGap.lean

GramObstruction.lean - new checked proof module

Current: https://github.com/alectionary/vibe-math/blob/master/formalization/stopping-bound/stronger/GramObstruction.lean

Immutable: https://github.com/alectionary/vibe-math/blob/bed7719cb3cac4118e4f333690db364566ede03e/formalization/stopping-bound/stronger/GramObstruction.lean

GrowingFamily.lean - new checked proof module

Current: https://github.com/alectionary/vibe-math/blob/master/formalization/stopping-bound/stronger/GrowingFamily.lean

Immutable: https://github.com/alectionary/vibe-math/blob/bed7719cb3cac4118e4f333690db364566ede03e/formalization/stopping-bound/stronger/GrowingFamily.lean

JointKernel.lean - new checked proof module

Current: https://github.com/alectionary/vibe-math/blob/master/formalization/stopping-bound/stronger/JointKernel.lean

Immutable: https://github.com/alectionary/vibe-math/blob/bed7719cb3cac4118e4f333690db364566ede03e/formalization/stopping-bound/stronger/JointKernel.lean

JointMatrices.lean - new checked proof module

Current: https://github.com/alectionary/vibe-math/blob/master/formalization/stopping-bound/stronger/JointMatrices.lean

Immutable: https://github.com/alectionary/vibe-math/blob/bed7719cb3cac4118e4f333690db364566ede03e/formalization/stopping-bound/stronger/JointMatrices.lean

Appendix B. Full Lean URLs (continued)

JointMomentFamily.lean - new checked proof module

Current: https://github.com/alectionary/vibe-math/blob/master/formalization/stopping-bound/stronger/JointMomentFamily.lean

Immutable: https://github.com/alectionary/vibe-math/blob/bed7719cb3cac4118e4f333690db364566ede03e/formalization/stopping-bound/stronger/JointMomentFamily.lean

JointPowers.lean - new checked proof module

Current: https://github.com/alectionary/vibe-math/blob/master/formalization/stopping-bound/stronger/JointPowers.lean

Immutable: https://github.com/alectionary/vibe-math/blob/bed7719cb3cac4118e4f333690db364566ede03e/formalization/stopping-bound/stronger/JointPowers.lean

JointSidon.lean - new checked proof module

Current: https://github.com/alectionary/vibe-math/blob/master/formalization/stopping-bound/stronger/JointSidon.lean

Immutable: https://github.com/alectionary/vibe-math/blob/bed7719cb3cac4118e4f333690db364566ede03e/formalization/stopping-bound/stronger/JointSidon.lean

JointSpectralBase.lean - new checked proof module

Current: https://github.com/alectionary/vibe-math/blob/master/formalization/stopping-bound/stronger/JointSpectralBase.lean

Immutable: https://github.com/alectionary/vibe-math/blob/bed7719cb3cac4118e4f333690db364566ede03e/formalization/stopping-bound/stronger/JointSpectralBase.lean

KernelFamily.lean - new checked proof module

Current: https://github.com/alectionary/vibe-math/blob/master/formalization/stopping-bound/stronger/KernelFamily.lean

Immutable: https://github.com/alectionary/vibe-math/blob/bed7719cb3cac4118e4f333690db364566ede03e/formalization/stopping-bound/stronger/KernelFamily.lean

KernelStopping.lean - new checked proof module

Current: https://github.com/alectionary/vibe-math/blob/master/formalization/stopping-bound/stronger/KernelStopping.lean

Immutable: https://github.com/alectionary/vibe-math/blob/bed7719cb3cac4118e4f333690db364566ede03e/formalization/stopping-bound/stronger/KernelStopping.lean

MomentBound.lean - new checked proof module

Current: https://github.com/alectionary/vibe-math/blob/master/formalization/stopping-bound/stronger/MomentBound.lean

Immutable: https://github.com/alectionary/vibe-math/blob/bed7719cb3cac4118e4f333690db364566ede03e/formalization/stopping-bound/stronger/MomentBound.lean

MomentExcess.lean - new checked proof module

Current: https://github.com/alectionary/vibe-math/blob/master/formalization/stopping-bound/stronger/MomentExcess.lean

Immutable: https://github.com/alectionary/vibe-math/blob/bed7719cb3cac4118e4f333690db364566ede03e/formalization/stopping-bound/stronger/MomentExcess.lean

Appendix B. Full Lean URLs (continued)

MomentReplacement.lean - new checked proof module

Current: https://github.com/alectionary/vibe-math/blob/master/formalization/stopping-bound/stronger/MomentReplacement.lean

Immutable: https://github.com/alectionary/vibe-math/blob/bed7719cb3cac4118e4f333690db364566ede03e/formalization/stopping-bound/stronger/MomentReplacement.lean

PartitionBound.lean - new checked proof module

Current: https://github.com/alectionary/vibe-math/blob/master/formalization/stopping-bound/stronger/PartitionBound.lean

Immutable: https://github.com/alectionary/vibe-math/blob/bed7719cb3cac4118e4f333690db364566ede03e/formalization/stopping-bound/stronger/PartitionBound.lean

PowersFamily.lean - new checked proof module

Current: https://github.com/alectionary/vibe-math/blob/master/formalization/stopping-bound/stronger/PowersFamily.lean

Immutable: https://github.com/alectionary/vibe-math/blob/bed7719cb3cac4118e4f333690db364566ede03e/formalization/stopping-bound/stronger/PowersFamily.lean

PowersThree.lean - new checked proof module

Current: https://github.com/alectionary/vibe-math/blob/master/formalization/stopping-bound/stronger/PowersThree.lean

Immutable: https://github.com/alectionary/vibe-math/blob/bed7719cb3cac4118e4f333690db364566ede03e/formalization/stopping-bound/stronger/PowersThree.lean

Practical27.lean - new checked proof module

Current: https://github.com/alectionary/vibe-math/blob/master/formalization/stopping-bound/stronger/Practical27.lean

Immutable: https://github.com/alectionary/vibe-math/blob/bed7719cb3cac4118e4f333690db364566ede03e/formalization/stopping-bound/stronger/Practical27.lean

Practical27Core.lean - new checked proof module

Current: https://github.com/alectionary/vibe-math/blob/master/formalization/stopping-bound/stronger/Practical27Core.lean

Immutable: https://github.com/alectionary/vibe-math/blob/bed7719cb3cac4118e4f333690db364566ede03e/formalization/stopping-bound/stronger/Practical27Core.lean

Practical27Design.lean - new checked proof module

Current: https://github.com/alectionary/vibe-math/blob/master/formalization/stopping-bound/stronger/Practical27Design.lean

Immutable: https://github.com/alectionary/vibe-math/blob/bed7719cb3cac4118e4f333690db364566ede03e/formalization/stopping-bound/stronger/Practical27Design.lean

Practical27Spectral.lean - new checked proof module

Current: https://github.com/alectionary/vibe-math/blob/master/formalization/stopping-bound/stronger/Practical27Spectral.lean

Immutable: https://github.com/alectionary/vibe-math/blob/bed7719cb3cac4118e4f333690db364566ede03e/formalization/stopping-bound/stronger/Practical27Spectral.lean

Appendix B. Full Lean URLs (continued)

RealGramBound.lean - new checked proof module

Current: https://github.com/alectionary/vibe-math/blob/master/formalization/stopping-bound/stronger/RealGramBound.lean

Immutable: https://github.com/alectionary/vibe-math/blob/bed7719cb3cac4118e4f333690db364566ede03e/formalization/stopping-bound/stronger/RealGramBound.lean

SeparateSidon.lean - new checked proof module

Current: https://github.com/alectionary/vibe-math/blob/master/formalization/stopping-bound/stronger/SeparateSidon.lean

Immutable: https://github.com/alectionary/vibe-math/blob/bed7719cb3cac4118e4f333690db364566ede03e/formalization/stopping-bound/stronger/SeparateSidon.lean

ShortFamily.lean - new checked proof module

Current: https://github.com/alectionary/vibe-math/blob/master/formalization/stopping-bound/stronger/ShortFamily.lean

Immutable: https://github.com/alectionary/vibe-math/blob/bed7719cb3cac4118e4f333690db364566ede03e/formalization/stopping-bound/stronger/ShortFamily.lean

ShortFamilySupports.lean - new checked proof module

Current: https://github.com/alectionary/vibe-math/blob/master/formalization/stopping-bound/stronger/ShortFamilySupports.lean

Immutable: https://github.com/alectionary/vibe-math/blob/bed7719cb3cac4118e4f333690db364566ede03e/formalization/stopping-bound/stronger/ShortFamilySupports.lean

SidonBad.lean - new checked proof module

Current: https://github.com/alectionary/vibe-math/blob/master/formalization/stopping-bound/stronger/SidonBad.lean

Immutable: https://github.com/alectionary/vibe-math/blob/bed7719cb3cac4118e4f333690db364566ede03e/formalization/stopping-bound/stronger/SidonBad.lean

SidonMatrices.lean - new checked proof module

Current: https://github.com/alectionary/vibe-math/blob/master/formalization/stopping-bound/stronger/SidonMatrices.lean

Immutable: https://github.com/alectionary/vibe-math/blob/bed7719cb3cac4118e4f333690db364566ede03e/formalization/stopping-bound/stronger/SidonMatrices.lean

SidonStopping.lean - new checked proof module

Current: https://github.com/alectionary/vibe-math/blob/master/formalization/stopping-bound/stronger/SidonStopping.lean

Immutable: https://github.com/alectionary/vibe-math/blob/bed7719cb3cac4118e4f333690db364566ede03e/formalization/stopping-bound/stronger/SidonStopping.lean

SourceConsumers.lean - new checked proof module

Current: https://github.com/alectionary/vibe-math/blob/master/formalization/stopping-bound/stronger/SourceConsumers.lean

Immutable: https://github.com/alectionary/vibe-math/blob/bed7719cb3cac4118e4f333690db364566ede03e/formalization/stopping-bound/stronger/SourceConsumers.lean

Appendix B. Full Lean URLs (continued)

SpectralMoment.lean - new checked proof module

Current: https://github.com/alectionary/vibe-math/blob/master/formalization/stopping-bound/stronger/SpectralMoment.lean

Immutable: https://github.com/alectionary/vibe-math/blob/bed7719cb3cac4118e4f333690db364566ede03e/formalization/stopping-bound/stronger/SpectralMoment.lean

SpectralPlane.lean - new checked proof module

Current: https://github.com/alectionary/vibe-math/blob/master/formalization/stopping-bound/stronger/SpectralPlane.lean

Immutable: https://github.com/alectionary/vibe-math/blob/bed7719cb3cac4118e4f333690db364566ede03e/formalization/stopping-bound/stronger/SpectralPlane.lean

StrongerAnswer.lean - new checked proof module

Current: https://github.com/alectionary/vibe-math/blob/master/formalization/stopping-bound/stronger/StrongerAnswer.lean

Immutable: https://github.com/alectionary/vibe-math/blob/bed7719cb3cac4118e4f333690db364566ede03e/formalization/stopping-bound/stronger/StrongerAnswer.lean

dependencies/IncidenceCore.lean - inherited proof dependency

Current:
https://github.com/alectionary/vibe-math/blob/master/formalization/stopping-bound/stronger/dependencies/IncidenceCore.lean

Immutable: https://github.com/alectionary/vibe-math/blob/bed7719cb3cac4118e4f333690db364566ede03e/formalization/stopping-bound/stronger/dependencies/IncidenceCore.lean

dependencies/IncidenceCover.lean - inherited proof dependency

Current:
https://github.com/alectionary/vibe-math/blob/master/formalization/stopping-bound/stronger/dependencies/IncidenceCover.lean

Immutable: https://github.com/alectionary/vibe-math/blob/bed7719cb3cac4118e4f333690db364566ede03e/formalization/stopping-bound/stronger/dependencies/IncidenceCover.lean

dependencies/IncidenceList.lean - inherited proof dependency

Current:
https://github.com/alectionary/vibe-math/blob/master/formalization/stopping-bound/stronger/dependencies/IncidenceList.lean

Immutable: https://github.com/alectionary/vibe-math/blob/bed7719cb3cac4118e4f333690db364566ede03e/formalization/stopping-bound/stronger/dependencies/IncidenceList.lean

dependencies/Spectral.lean - inherited proof dependency

Current: https://github.com/alectionary/vibe-math/blob/master/formalization/stopping-bound/stronger/dependencies/Spectral.lean

Immutable: https://github.com/alectionary/vibe-math/blob/bed7719cb3cac4118e4f333690db364566ede03e/formalization/stopping-bound/stronger/dependencies/Spectral.lean

Appendix B. Cited earlier proof closure

The42 historical files below support the cited original-parameter and B1 results. Earlier then-open replacement labels are superseded by this paper's positive existence answer. Their unchanged publication manifests and immutable sources are retained. These are cited prior proofs, not new declarations in this paper.

formalization/stopping-bound/Audit.lean

Current: https://github.com/alectionary/vibe-math/blob/master/formalization/stopping-bound/Audit.lean

Immutable: https://github.com/alectionary/vibe-math/blob/b7628643bb256bf9c1a319f16f6e19608116cff6/formalization/stopping-bound/Audit.lean

formalization/stopping-bound/Spectral.lean

Current: https://github.com/alectionary/vibe-math/blob/master/formalization/stopping-bound/Spectral.lean

Immutable: https://github.com/alectionary/vibe-math/blob/b7628643bb256bf9c1a319f16f6e19608116cff6/formalization/stopping-bound/Spectral.lean

formalization/stopping-bound/classification/AuditClassification.lean

Current: https://github.com/alectionary/vibe-math/blob/master/formalization/stopping-bound/classification/AuditClassification.lean

Immutable: https://github.com/alectionary/vibe-math/blob/b7628643bb256bf9c1a319f16f6e19608116cff6/formalization/stopping-bound/classification/AuditClassification.lean

formalization/stopping-bound/classification/Classification.lean

Current: https://github.com/alectionary/vibe-math/blob/master/formalization/stopping-bound/classification/Classification.lean

Immutable: https://github.com/alectionary/vibe-math/blob/b7628643bb256bf9c1a319f16f6e19608116cff6/formalization/stopping-bound/classification/Classification.lean

formalization/stopping-bound/classification/ClassificationBranches.lean

Current:
https://github.com/alectionary/vibe-math/blob/master/formalization/stopping-bound/classification/ClassificationBranches.lean

Immutable: https://github.com/alectionary/vibe-math/blob/b7628643bb256bf9c1a319f16f6e19608116cff6/formalization/stopping-bound/classification/ClassificationBranches.lean

formalization/stopping-bound/classification/ClassificationCertificate.lean

Current:
https://github.com/alectionary/vibe-math/blob/master/formalization/stopping-bound/classification/ClassificationCertificate.lean

Immutable: https://github.com/alectionary/vibe-math/blob/b7628643bb256bf9c1a319f16f6e19608116cff6/formalization/stopping-bound/classification/ClassificationCertificate.lean

formalization/stopping-bound/classification/ClassificationData.lean

Current: https://github.com/alectionary/vibe-math/blob/master/formalization/stopping-bound/classification/ClassificationData.lean

Immutable: https://github.com/alectionary/vibe-math/blob/b7628643bb256bf9c1a319f16f6e19608116cff6/formalization/stopping-bound/classification/ClassificationData.lean

formalization/stopping-bound/classification/ClassificationKernel.lean

Current:
https://github.com/alectionary/vibe-math/blob/master/formalization/stopping-bound/classification/ClassificationKernel.lean

Immutable: https://github.com/alectionary/vibe-math/blob/b7628643bb256bf9c1a319f16f6e19608116cff6/formalization/stopping-bound/classification/ClassificationKernel.lean

Appendix B. Cited earlier proof closure (continued)

formalization/stopping-bound/classification/ClassificationSearch.lean

Current:
https://github.com/alectionary/vibe-math/blob/master/formalization/stopping-bound/classification/ClassificationSearch.lean

Immutable: https://github.com/alectionary/vibe-math/blob/b7628643bb256bf9c1a319f16f6e19608116cff6/formalization/stopping-bound/classification/ClassificationSearch.lean

formalization/stopping-bound/classification/dependencies/Incidence.lean

Current:
https://github.com/alectionary/vibe-math/blob/master/formalization/stopping-bound/classification/dependencies/Incidence.lean

Immutable: https://github.com/alectionary/vibe-math/blob/b7628643bb256bf9c1a319f16f6e19608116cff6/formalization/stopping-bound/classification/dependencies/Incidence.lean

formalization/stopping-bound/classification/dependencies/IncidenceCSS.lean

Current:
https://github.com/alectionary/vibe-math/blob/master/formalization/stopping-bound/classification/dependencies/IncidenceCSS.lean

Immutable: https://github.com/alectionary/vibe-math/blob/b7628643bb256bf9c1a319f16f6e19608116cff6/formalization/stopping-bound/classification/dependencies/IncidenceCSS.lean

formalization/stopping-bound/classification/dependencies/IncidenceCertificate.lean

Current: https://github.com/alectionary/vibe-math/blob/master/formalization/stopping-bound/classification/dependencies/IncidenceCertificate.lean

Immutable: https://github.com/alectionary/vibe-math/blob/b7628643bb256bf9c1a319f16f6e19608116cff6/formalization/stopping-bound/classification/dependencies/IncidenceCertificate.lean

formalization/stopping-bound/classification/dependencies/IncidenceCore.lean

Current: https://github.com/alectionary/vibe-math/blob/master/formalization/stopping-bound/classification/dependencies/IncidenceCore.lean

Immutable: https://github.com/alectionary/vibe-math/blob/b7628643bb256bf9c1a319f16f6e19608116cff6/formalization/stopping-bound/classification/dependencies/IncidenceCore.lean

formalization/stopping-bound/classification/dependencies/IncidenceCover.lean

Current: https://github.com/alectionary/vibe-math/blob/master/formalization/stopping-bound/classification/dependencies/IncidenceCover.lean

Immutable: https://github.com/alectionary/vibe-math/blob/b7628643bb256bf9c1a319f16f6e19608116cff6/formalization/stopping-bound/classification/dependencies/IncidenceCover.lean

formalization/stopping-bound/classification/dependencies/IncidenceDegrees.lean

Current: https://github.com/alectionary/vibe-math/blob/master/formalization/stopping-bound/classification/dependencies/IncidenceDegrees.lean

Immutable: https://github.com/alectionary/vibe-math/blob/b7628643bb256bf9c1a319f16f6e19608116cff6/formalization/stopping-bound/classification/dependencies/IncidenceDegrees.lean

formalization/stopping-bound/classification/dependencies/IncidenceFamily.lean

Current: https://github.com/alectionary/vibe-math/blob/master/formalization/stopping-bound/classification/dependencies/IncidenceFamily.lean

Immutable: https://github.com/alectionary/vibe-math/blob/b7628643bb256bf9c1a319f16f6e19608116cff6/formalization/stopping-bound/classification/dependencies/IncidenceFamily.lean

Appendix B. Cited earlier proof closure (continued)

formalization/stopping-bound/classification/dependencies/IncidenceFamilyCore.lean

Current: https://github.com/alectionary/vibe-math/blob/master/formalization/stopping-bound/classification/dependencies/IncidenceFamilyCore.lean

Immutable: https://github.com/alectionary/vibe-math/blob/b7628643bb256bf9c1a319f16f6e19608116cff6/formalization/stopping-bound/classification/dependencies/IncidenceFamilyCore.lean

formalization/stopping-bound/classification/dependencies/IncidenceFamilySource.lean

Current: https://github.com/alectionary/vibe-math/blob/master/formalization/stopping-bound/classification/dependencies/IncidenceFamilySource.lean

Immutable: https://github.com/alectionary/vibe-math/blob/b7628643bb256bf9c1a319f16f6e19608116cff6/formalization/stopping-bound/classification/dependencies/IncidenceFamilySource.lean

formalization/stopping-bound/classification/dependencies/IncidenceList.lean

Current: https://github.com/alectionary/vibe-math/blob/master/formalization/stopping-bound/classification/dependencies/IncidenceList.lean

Immutable: https://github.com/alectionary/vibe-math/blob/b7628643bb256bf9c1a319f16f6e19608116cff6/formalization/stopping-bound/classification/dependencies/IncidenceList.lean

formalization/stopping-bound/classification/dependencies/IncidenceSource.lean

Current: https://github.com/alectionary/vibe-math/blob/master/formalization/stopping-bound/classification/dependencies/IncidenceSource.lean

Immutable: https://github.com/alectionary/vibe-math/blob/b7628643bb256bf9c1a319f16f6e19608116cff6/formalization/stopping-bound/classification/dependencies/IncidenceSource.lean

formalization/stopping-bound/classification/dependencies/IncidenceSpectral.lean

Current: https://github.com/alectionary/vibe-math/blob/master/formalization/stopping-bound/classification/dependencies/IncidenceSpectral.lean

Immutable: https://github.com/alectionary/vibe-math/blob/b7628643bb256bf9c1a319f16f6e19608116cff6/formalization/stopping-bound/classification/dependencies/IncidenceSpectral.lean

formalization/stopping-bound/classification/dependencies/Spectral.lean

Current: https://github.com/alectionary/vibe-math/blob/master/formalization/stopping-bound/classification/dependencies/Spectral.lean

Immutable: https://github.com/alectionary/vibe-math/blob/b7628643bb256bf9c1a319f16f6e19608116cff6/formalization/stopping-bound/classification/dependencies/Spectral.lean

formalization/stopping-bound/incidence/AuditIncidence.lean

Current: https://github.com/alectionary/vibe-math/blob/master/formalization/stopping-bound/incidence/AuditIncidence.lean

Immutable: https://github.com/alectionary/vibe-math/blob/b7628643bb256bf9c1a319f16f6e19608116cff6/formalization/stopping-bound/incidence/AuditIncidence.lean

formalization/stopping-bound/incidence/Incidence.lean

Current: https://github.com/alectionary/vibe-math/blob/master/formalization/stopping-bound/incidence/Incidence.lean

Immutable: https://github.com/alectionary/vibe-math/blob/b7628643bb256bf9c1a319f16f6e19608116cff6/formalization/stopping-bound/incidence/Incidence.lean

Appendix B. Cited earlier proof closure (continued)

formalization/stopping-bound/incidence/IncidenceCSS.lean

Current: https://github.com/alectionary/vibe-math/blob/master/formalization/stopping-bound/incidence/IncidenceCSS.lean

Immutable: https://github.com/alectionary/vibe-math/blob/b7628643bb256bf9c1a319f16f6e19608116cff6/formalization/stopping-bound/incidence/IncidenceCSS.lean

formalization/stopping-bound/incidence/IncidenceCertificate.lean

Current: https://github.com/alectionary/vibe-math/blob/master/formalization/stopping-bound/incidence/IncidenceCertificate.lean

Immutable: https://github.com/alectionary/vibe-math/blob/b7628643bb256bf9c1a319f16f6e19608116cff6/formalization/stopping-bound/incidence/IncidenceCertificate.lean

formalization/stopping-bound/incidence/IncidenceCore.lean

Current: https://github.com/alectionary/vibe-math/blob/master/formalization/stopping-bound/incidence/IncidenceCore.lean

Immutable: https://github.com/alectionary/vibe-math/blob/b7628643bb256bf9c1a319f16f6e19608116cff6/formalization/stopping-bound/incidence/IncidenceCore.lean

formalization/stopping-bound/incidence/IncidenceCover.lean

Current: https://github.com/alectionary/vibe-math/blob/master/formalization/stopping-bound/incidence/IncidenceCover.lean

Immutable: https://github.com/alectionary/vibe-math/blob/b7628643bb256bf9c1a319f16f6e19608116cff6/formalization/stopping-bound/incidence/IncidenceCover.lean

formalization/stopping-bound/incidence/IncidenceDegrees.lean

Current: https://github.com/alectionary/vibe-math/blob/master/formalization/stopping-bound/incidence/IncidenceDegrees.lean

Immutable: https://github.com/alectionary/vibe-math/blob/b7628643bb256bf9c1a319f16f6e19608116cff6/formalization/stopping-bound/incidence/IncidenceDegrees.lean

formalization/stopping-bound/incidence/IncidenceFamily.lean

Current: https://github.com/alectionary/vibe-math/blob/master/formalization/stopping-bound/incidence/IncidenceFamily.lean

Immutable: https://github.com/alectionary/vibe-math/blob/b7628643bb256bf9c1a319f16f6e19608116cff6/formalization/stopping-bound/incidence/IncidenceFamily.lean

formalization/stopping-bound/incidence/IncidenceFamilyCore.lean

Current: https://github.com/alectionary/vibe-math/blob/master/formalization/stopping-bound/incidence/IncidenceFamilyCore.lean

Immutable: https://github.com/alectionary/vibe-math/blob/b7628643bb256bf9c1a319f16f6e19608116cff6/formalization/stopping-bound/incidence/IncidenceFamilyCore.lean

formalization/stopping-bound/incidence/IncidenceFamilySource.lean

Current:
https://github.com/alectionary/vibe-math/blob/master/formalization/stopping-bound/incidence/IncidenceFamilySource.lean

Immutable: https://github.com/alectionary/vibe-math/blob/b7628643bb256bf9c1a319f16f6e19608116cff6/formalization/stopping-bound/incidence/IncidenceFamilySource.lean

Appendix B. Cited earlier proof closure (continued)

formalization/stopping-bound/incidence/IncidenceList.lean

Current: https://github.com/alectionary/vibe-math/blob/master/formalization/stopping-bound/incidence/IncidenceList.lean

Immutable: https://github.com/alectionary/vibe-math/blob/b7628643bb256bf9c1a319f16f6e19608116cff6/formalization/stopping-bound/incidence/IncidenceList.lean

formalization/stopping-bound/incidence/IncidenceSource.lean

Current: https://github.com/alectionary/vibe-math/blob/master/formalization/stopping-bound/incidence/IncidenceSource.lean

Immutable: https://github.com/alectionary/vibe-math/blob/b7628643bb256bf9c1a319f16f6e19608116cff6/formalization/stopping-bound/incidence/IncidenceSource.lean

formalization/stopping-bound/incidence/IncidenceSpectral.lean

Current: https://github.com/alectionary/vibe-math/blob/master/formalization/stopping-bound/incidence/IncidenceSpectral.lean

Immutable: https://github.com/alectionary/vibe-math/blob/b7628643bb256bf9c1a319f16f6e19608116cff6/formalization/stopping-bound/incidence/IncidenceSpectral.lean

formalization/stopping-bound/incidence/dependencies/Spectral.lean

Current:
https://github.com/alectionary/vibe-math/blob/master/formalization/stopping-bound/incidence/dependencies/Spectral.lean

Immutable: https://github.com/alectionary/vibe-math/blob/b7628643bb256bf9c1a319f16f6e19608116cff6/formalization/stopping-bound/incidence/dependencies/Spectral.lean

formalization/stopping-bound/source-answer/AuditDenseSourceBridge.lean

Current:
https://github.com/alectionary/vibe-math/blob/master/formalization/stopping-bound/source-answer/AuditDenseSourceBridge.lean

Immutable: https://github.com/alectionary/vibe-math/blob/b7628643bb256bf9c1a319f16f6e19608116cff6/formalization/stopping-bound/source-answer/AuditDenseSourceBridge.lean

formalization/stopping-bound/source-answer/AuditSourceAnswer.lean

Current:
https://github.com/alectionary/vibe-math/blob/master/formalization/stopping-bound/source-answer/AuditSourceAnswer.lean

Immutable: https://github.com/alectionary/vibe-math/blob/b7628643bb256bf9c1a319f16f6e19608116cff6/formalization/stopping-bound/source-answer/AuditSourceAnswer.lean

formalization/stopping-bound/source-answer/DenseSourceBridge.lean

Current:
https://github.com/alectionary/vibe-math/blob/master/formalization/stopping-bound/source-answer/DenseSourceBridge.lean

Immutable: https://github.com/alectionary/vibe-math/blob/b7628643bb256bf9c1a319f16f6e19608116cff6/formalization/stopping-bound/source-answer/DenseSourceBridge.lean

formalization/stopping-bound/source-answer/SourceQuestionAnswer.lean

Current:
https://github.com/alectionary/vibe-math/blob/master/formalization/stopping-bound/source-answer/SourceQuestionAnswer.lean

Immutable: https://github.com/alectionary/vibe-math/blob/b7628643bb256bf9c1a319f16f6e19608116cff6/formalization/stopping-bound/source-answer/SourceQuestionAnswer.lean

Appendix B. Cited earlier proof closure (continued)

formalization/stopping-bound/source-answer/dependencies/Spectral.lean

Current:
https://github.com/alectionary/vibe-math/blob/master/formalization/stopping-bound/source-answer/dependencies/Spectral.lean

Immutable: https://github.com/alectionary/vibe-math/blob/b7628643bb256bf9c1a319f16f6e19608116cff6/formalization/stopping-bound/source-answer/dependencies/Spectral.lean

formalization/stopping-bound/source-answer/source-ledgers/parent-Audit.lean

Current: https://github.com/alectionary/vibe-math/blob/master/formalization/stopping-bound/source-answer/source-ledgers/parent-Audit.lean

Immutable: https://github.com/alectionary/vibe-math/blob/b7628643bb256bf9c1a319f16f6e19608116cff6/formalization/stopping-bound/source-answer/source-ledgers/parent-Audit.lean

Appendix C. Source chronology and explicit calendar deltas

This manuscript uses24October2026 inAmerica/Los_Angeles as its dated revision. UTC research/build times are recorded separately. Civil calendar days to 2026-10-24; exact dates give exact differences. Month/year precision yields inclusive minimum/maximum day ranges; dates are not invented. Date ranges are clipped to the revision because these records already existed then. Source age is not unresolved-problem duration.

The ledger includes inherited comparison sources, all newly used primary papers, public software snapshots and held discovery leads. A held lead has an inspection-only delta when its earliest date was not assessed. This does not make the inspection its publication date. Source age is not an unsolved duration.

Source / role	Milestone, date and calendar days	Access / precision
F16: Abiad-Castriota: motivating parameter question; Original question; Prop.5.1 / Remark5.2 / §6	first and inspected v1: 2026-07-15; 101 days (day)	inspected full text
F68: Tanner: Minimum Distance Bounds by Graph Analysis; Classical spectral ingredient	earliest inspected manuscript: 2000-07-26; 9586 days (day) journal milestone: 2001; 9063-9427 days (year)	inspected full text
F67: Zhu-Chugg: Bounds on the Expansion Properties of Tanner Graphs; Classical stopping bound; exact Cor.6 comparison	conference predecessor metadata: 2005-09-28; 7696 days (day) inspected journal version: 2007-12; 6872-6902 days (month)	inspected full text
F69: Panteleev-Kalachev: Degenerate Quantum LDPC Codes With Good Finite Length Performance; Scalar-factor construction	first v1: 2019-04-04; 2760 days (day) inspected v3: 2021-11-12; 1807 days (day) journal milestone: 2021-11-22; 1797 days (day)	inspected full text
F70: Leverrier-Zemor: Quantum Tanner codes; Constituent Ramanujan comparison	first v1: 2022-02-28; 1699 days (day) inspected v3: 2022-09-16; 1499 days (day) PDF internal date: 2022-09-19; 1496 days (day)	inspected full text
F71: Feng-Li: Spectra of Hypergraphs and Applications; Classical source cited by Remark5.2; not a proof premise	publication metadata: 1996-09; 10981-11010 days (month)	abstract only full text gap
F72: Li-Sole: Spectra of Regular Graphs and Hypergraphs and Orthogonal Polynomials; Classical comparison lead; not a proof premise	publication metadata: 1996; 10889-11254 days (year)	partial web text archival gap
F73: Prabhu et al.: On the Efficacy of the Peeling Decoder for the Quantum Expander Code; Known stopping-set mechanisms and expansion conditions	first v1: 2025-04-30; 542 days (day) inspected v2: 2025-05-29; 513 days (day)	inspected full text
F74: Wang-Kulkarni-Poor: Finding All Small Error-Prone Substructures in LDPC Codes; Classical rooted narrowing and branch-and-bound	related first preprint: 2006-05-19; 7463 days (day) inspected current-version date: 2009-04-22; 6394 days (day) journal issue: 2009-05; 6355-6385 days (month)	inspected full text
F75: Yao-Gokduman-Pfister: Cluster Decomposition for Improved Erasure Decoding of Quantum LDPC Codes; B1 simulations / peeling-cluster comparison	first and inspected v1: 2024-12-11; 682 days (day)	inspected full text

Source / role	Milestone, date and calendar days	Access / precision
Connolly2024: Connolly-Londe-Leverrier-Delfosse: Fast erasure decoder for hypergraph product codes; Original stabilizer-support mechanism, credited through inspected F73/F75	first v1: 2022-08-01; 1545 days (day) v3 metadata: 2024-08-21; 794 days (day) journal publication: 2024-08-27; 788 days (day)	metadata only full text not consulted
F76: Prabhu-Gayathri-Kumar: On the Analysis of Stopping Sets Associated with the Quantum Hypergraph Product Code; Later comparison lead; no exact theorem claim	conference metadata: 2026; 0-296 days (year)	full text access gap
RosnesYtrehus2007: Rosnes-Ytrehus: An Algorithm to Find All Small-Size Stopping Sets of Low-Density Parity-Check Matrices; Bibliography-only enumeration lead	conference year: 2007; 6872-7236 days (year)	bibliography only not used as theorem evidence
Morris2025: Kirsten Morris: Analysis of Graph-based Decoders for Quantum Low Density Parity Check Codes; Discovery trail through existing dissertation inventory	repository source version: 2025-08; 419-449 days (month)	discovery record not new full text screen
qLDPCOrg/qLDPC: qLDPCOrg/qLDPC pinned snapshot; Software/infrastructure snapshot, not original publication age	inspected commit: 2026-10-23; 1 days (day)	pinned source or infrastructure
Lcrypto/Stopping-sets-enumerator: Lcrypto/Stopping-sets-enumerator pinned snapshot; Software/infrastructure snapshot, not original publication age	inspected commit: 2023-10-29; 1091 days (day)	pinned source or infrastructure
leanprover-community/mathlib4: leanprover-community/mathlib4 pinned snapshot; Software/infrastructure snapshot, not original publication age	inspected commit: 2026-10-21; 3 days (day)	pinned source or infrastructure
X-source-announcement: mathCObot source-paper announcement; Accessible X comparison; not a mathematical answer	post UTC date: 2026-07-16; 100 days (day) authenticated browser local displayed post date: 2026-07-15; 101 days (day)	authenticated post detail rechecked final checkpoint
X-earlier-result: A_Lectionary earlier absorbing-set announcement; Earlier repository result; different target	post UTC date: 2026-10-24; 0 days (day)	post inspected in parent audit
other-code-snapshots: Other inspected GitHub mirrors/README-only code; Secondary discovery/control sources; no theorem or earliest-publication claim	recorded inspection: 2026-10-24; 0 days (day)	inspection date only earliest dates unassessed
Lean-runtime: Lean4.36.0-rc1 runtime; Pinned proof checker; release-age claim not made	recorded use: 2026-10-24; 0 days (day)	use date only release date unassessed

Source / role	Milestone, date and calendar days	Access / precision
F77: Schwartz-Vardy: stopping distance and stopping redundancy; Classical matrix-dependent stopping convention, codeword mechanism and distinct-column criterion	first preprint metadata: 2005-03-23; 7885 days (day) journal revised manuscript: 2005-12-09; 7624 days (day) inspected journal issue: 2006-03; 7512-7542 days (month)	inspected full text
F78: Wang-Pryadko: Distance bounds for generalized bicycle codes; Binary GB/CSS construction convention and logical-distance comparison	first and inspected v1 submission: 2022-03-31; 1668 days (day) PDF internal date: 2022-04-01; 1667 days (day)	inspected full text
F79: Kuo-Ouyang: Degenerate quantum erasure decoding; Quantum-aware decoding comparison; article-in-press main text screened	first preprint metadata: 2024-11-20; 703 days (day) received: 2025-05-31; 511 days (day) accepted: 2026-02-23; 243 days (day) publication / inspected article-in-press: 2026-03-13; 225 days (day) version of record metadata: 2026-04-29; 178 days (day)	inspected full text
F80: Kashyap-Vardy: Stopping Sets in Codes from Designs; Classical uniform-overlap lower bound, Theorem1	reported conference presentation: 2003; 8333-8697 days (year) reported original composition: 2003-09; 8425-8454 days (month) inspected revised author manuscript: 2026-10-24; 0 days (day)	full web text screened direct pdf download403 transcription archived
F81: Xia-Fu: Minimum Pseudo-Weight and Minimum Pseudo-Codewords of LDPC Codes; Classical overlap/pseudo-weight comparison, Theorems1/3; full17-page primary screen	earliest preprint: 2006-06-12; 7439 days (day) inspected arXivv2 stamp: 2006-11-11; 7287 days (day) PDF internal title-page date: 2019-05-06; 2728 days (day)	inspected full text
F38: Dehghan-Banihashemi: Cospectral Bipartite Graphs with the Same Degree Sequences but with Different Number of Large Cycles; Earlier cospectral-cycle obstruction; existing23-page source screen reused, no added inventory cohort	first and inspectedv1: 2019-05-31; 2703 days (day) journal volume metadata: 2019; 2489-2853 days (year)	inspected full text
gideonlsx/bicycle-race: gideonlsx/bicycle-race pinned public repository; README/software comparison; not theorem evidence	inspected commit: 2026-10-11; 13 days (day)	public metadata or readme inspected
kywukuo/qLDPC_dec: kywukuo/qLDPC_dec pinned public repository; README/software comparison; not theorem evidence	inspected commit: 2026-01-26; 271 days (day)	public metadata or readme inspected
prior-proof-checkpoint: Cited original-question and B1 proof closure; Earlier checked proof sources, not a new completion	immutable proof-source commit: 2026-10-24; 0 days (day)	checked prior lean sources
new-proof-checkpoint: This stronger-bound proof checkpoint; Checked mathematical source, not a priority date	new research and proof-source checkpoint: 2026-10-24; 0 days (day)	checked lean source
held-lead-1: Generalized Bicycle Codes with Low Connectivity: Minimum Distance Bounds and Hook Errors; Encountered lead, no theorem premise or zero-question claim	inspection only: 2026-10-24; 0 days (day)	metadata or abstract only fulltext unassessed

Source / role	Milestone, date and calendar days	Access / precision
held-lead-2: Spectral Theory of Semisimple Bivariate Bicycle Codes; Encountered lead, no theorem premise or zero-question claim	inspection only: 2026-10-24; 0 days (day)	metadata or abstract only fulltext unassessed
held-lead-3: Logical Operator Decomposition for Distance Analysis of Bivariate Bicycle Codes; Encountered lead, no theorem premise or zero-question claim	inspection only: 2026-10-24; 0 days (day)	metadata or abstract only fulltext unassessed
held-lead-4: Permutation Decoding and the Stopping Redundancy Hierarchy; Encountered lead, no theorem premise or zero-question claim	inspection only: 2026-10-24; 0 days (day)	metadata or abstract only fulltext unassessed
held-lead-5: Quantum Maxwell Erasure Decoder for qLDPC codes; Encountered lead, no theorem premise or zero-question claim	inspection only: 2026-10-24; 0 days (day)	metadata or abstract only fulltext unassessed
held-lead-6: Stopping Sets of Algebraic Geometry Codes; Encountered lead, no theorem premise or zero-question claim	inspection only: 2026-10-24; 0 days (day)	metadata or abstract only fulltext unassessed
held-lead-7: On the Hardness of Approximating Stopping and Trapping Sets in LDPC Codes; Encountered lead, no theorem premise or zero-question claim	inspection only: 2026-10-24; 0 days (day)	metadata or abstract only fulltext unassessed
held-lead-8: A class of LDPC codes based on symplectic space over finite fields; Encountered lead, no theorem premise or zero-question claim	inspection only: 2026-10-24; 0 days (day)	metadata or abstract only fulltext unassessed
F82: Donovan-Rao-Üsküplü-Yazıcı: QC-LDPC Codes from Difference Matrices and Difference Covering Arrays; Post-lemma primary full-document question screen and exact construction comparison	first and inspectedv1: 2022-05-01; 1637 days (day) journal publication year metadata: 2023; 1028-1392 days (year)	inspected full text
F83: Chee-Ling-Yin: Optimal Partitioned Cyclic Difference Packings; Post-lemma primary full-document question screen and exact construction comparison	first and inspectedv1: 2010-08-10; 5919 days (day) journal issue metadata: 2010-11; 5807-5836 days (month)	inspected full text
F84: Gholami-Samadieh: QC-LDPC Codes Based on Finite Set Systems; Post-lemma primary full-document question screen and exact construction comparison	first and inspectedv1: 2014-05-15; 4545 days (day) internal PDF title-page date: 2022-03-28; 1671 days (day)	inspected full text

Source / role	Milestone, date and calendar days	Access / precision
F85: Amirzade-Sadeghi: Lower bounds on the lifting degree by difference matrices; Post-lemma primary full-document question screen and exact construction comparison	first and inspectedv1: 2017-09-04; 3337 days (day) internal PDF running draft date: 2018-11-18; 2897 days (day)	inspected full text
postlemma-held-1: Difference-Spectrum Completion of Multi-Weight Optical Orthogonal Codes; Post-lemma metadata lead, no mathematical premise or zero-question claim	inspection only: 2026-10-24; 0 days (day)	publisher429 and primary PDF403; fulltext unavailable; metadata only no zero question claim
postlemma-held-2: On the Girth of Tanner QC-LDPC Cycle Codes: An Algebraic Number Theory Approach; Post-lemma metadata lead, no mathematical premise or zero-question claim	inspection only: 2026-10-24; 0 days (day)	publisher429 and primary PDF403; fulltext unavailable; metadata only no zero question claim
postlemma-held-3: Cyclic balanced incomplete block designs with block size four and arbitrary index; Post-lemma metadata lead, no mathematical premise or zero-question claim	inspection only: 2026-10-24; 0 days (day)	publisher403; abstract metadata only; fulltext unavailable no zero question claim
postlemma-held-4: Additive, Structural, and Multiplicative Transformations for the Construction of Quasi-Cyclic LDPC Matrices; Post-lemma metadata lead, no mathematical premise or zero-question claim	inspection only: 2026-10-24; 0 days (day)	Secondary search snippet only; primary full version not located/inspected in bounded search; no theorem premise or zero-question claim
postlemma-held-5: Cyclic Difference Packing and Covering Arrays; Post-lemma metadata lead, no mathematical premise or zero-question claim	inspection only: 2026-10-24; 0 days (day)	Secondary metadata/ bibliographic lead only; primary full version not inspected; no theorem premise or zero-question claim
X-final-search: Final authenticated X source/stopping audit; Read-only bounded query with known-positive control; not absence or theorem evidence	final inspection: 2026-10-24; 0 days (day)	authenticated bounded search read no post sent

Inspection-only dates are explicitly distinguished from publication dates. The Kashyap-Vardy author manuscript is undated after its reported2003 composition; its web transcription hash is not a PDF hash. Supplementary/final-version, citation-index and X access gaps remain named. Date differences are source ages, not durations of unsolved problems.

Investigation block:24October22:59:20UTC to25October01:59:20UTC. First incidence/exact-six checked checkpoint recorded at 2026-10-24T23:36:01.519761+00:00 (0h 36m 41s elapsed). Final expanded checked checkpoint recorded at 2026-10-25T01:30:24.643550+00:00 (2h 31m 4s elapsed). Manuscript and Git publication followed separately. These are checked-result checkpoints; informal discovery times are not independently fixed.